cutoff on the allowed modes, we get a finite answer rather than an infinite one, but that answer is 10^{122} times larger than the value we actually observe. This mismatch, known as the *cosmological constant problem*, has often been called the biggest discrepancy between theory and observation in all of physics.

The cosmological constant problem is not really a conflict between theory and observation in the strict sense. We don't have anything like a reliable theoretical prediction for what the vacuum energy should be. Our very wrong estimate comes from making two dubious assumptions: that the classical contribution to the vacuum energy is zero, and that we impose a cutoff at the Planck scale. It's always possible that the classical contribution we should start with is almost exactly as large as the quantum piece, but with the opposite sign, so that when we add them together we get an observed "physical" vacuum energy with a relatively tiny value. We just have no idea why that should be true.

The problem is not that theory conflicts with observation; it's that our rough expectations are way off, which most people take as a clue that something mysterious and unknown is at work. Since the energy we estimated was a purely quantum-mechanical effect, and we measure its existence using its gravitational effect, it's plausible that we won't solve the problem until we have a fully working quantum theory of gravity.

Popular discussions of quantum field theory will often describe the vacuum as full of "quantum fluctuations," or even "particles popping in and out of existence in empty space." That's an evocative picture, but it's more false than true.

In empty space described by the quantum-field-theory vacuum, nothing is fluctuating at all; the quantum state is absolutely stationary. The picture of particles popping in and out of existence is entirely different from the reality, in which the state is precisely the same from one moment to another. There is undoubtedly an intrinsically quantum contribution to the energy of empty space, but it's misleading to speak of that energy as coming from "fluctuations," when nothing is actually fluctuating. The system is sitting peacefully in its lowest-energy quantum state.